The Media Connectivity Association



OVERVIEW AND UPDATES · CONNECTED MEDIA PRODUCTION

Non-Public Networks

Private networks replacing legacy COFDM and RF-based contribution links

3GPP

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About the project

The documents the project rests on

3GPP TS 22.263

Technical Specification

Service requirements for Video, Imaging and Audio for Professional Applications (VIAPA), the requirements most often cited for NPNs in broadcast use.

3GPP TS 23.501

Technical Specification

The system architecture of the 5G System. It defines the SNPN and PNI-NPN deployment models.

3GPP TS 33.501

Technical Specification

The security architecture and procedures of the 5G System, including NPN security.

FIND OUT MORE **Technical Analysis** **Specifications**

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Deployment models

A standalone NPN is independent; an integrated NPN reuses the public network of an operator

SNPN Stand-alone Non-Public Network

- Own 5G core
- Independent of any PLMN
- PLMN ID and NID
- Self-assigned or coordinated NID
- SNPN network selection by the UE

PNI-NPN Public Network Integrated NPN

- Realised through the operator's PLMN
- Part of the host PLMN
- Dedicated DNN
- Dedicated network slice (S-NSSAI)
- Closed Access Group (CAG)

In an SNPN the organisation runs the core, so it controls the UDM, PCF and SMF configuration directly. A CAG is advertised by cells; only UEs that are members may camp on and access them.

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Identity, access and production

Release 17 separated credentials from the network; production traffic is uplink-heavy

Identity and onboarding Release 17

- Credentials Holder
- External AAA server
- Onboarding SNPN (ON-SNPN)
- Default Credentials Server (DCS)
- Provisioning Server (PVS)

Access beyond the radio link

- Untrusted non-3GPP access
- Trusted non-3GPP access
- Wired LAN, Wi-Fi, fixed backhaul

QoS and slicing for media production

- Sustained uplink
- 5G QoS Flows and 5QI
- Guaranteed Bit Rate
- S-NSSAI per service class
- Time Sensitive Communications

Spectrum a national regulatory matter

- Locally licensed spectrum
- Spectrum leased from an operator
- Shared access
- Unlicensed bands

Spectrum bands, licence conditions and power limits differ between administrations and are checked with the national regulator. Access over non-3GPP networks is defined in TS 24.502.

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How it grew

The 3GPP work on non-public networks, release by release

Release 16

The baseline: SNPN and PNI-NPN, the NID and the CAG (Vertical_LAN, NG_RAN_PRN).

Release 17

Credentials Holder, external AAA authentication, UE onboarding and remote provisioning, IMS voice and emergency services and PWS for SNPN.

Release 18

Mobility between equivalent SNPNs, non-3GPP access for SNPN, and localised services via a hosting network (eNPN_Ph2).

Release 19

Security for a PLMN hosting an NPN, interconnect of SNPNs, ProSe in NPN and CAG-based access for NR Femto.

Release 20

Release 20 (work plan of 28 September 2026): security for a PLMN hosting an NPN, phase 2 (TS 33.501). Unverified: no TR 21.920 yet.

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